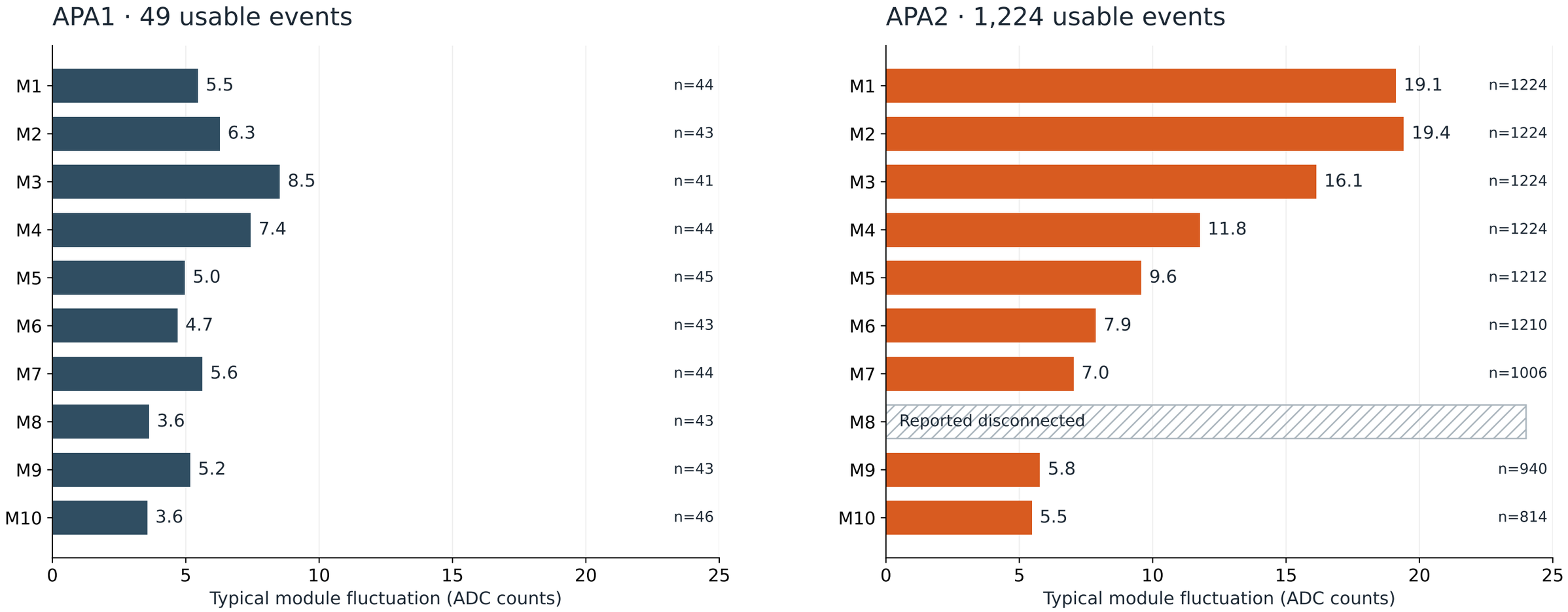
The precursor is strongest in the upper modules of both APAs

APA1 peaks in M3–M4; APA2 peaks in M1–M2. M1 is the top module and M10 the bottom module in these physical maps.



Within individual events: M3/M4 is largest in 39/49 usable APA1 events; M1/M2 is largest in 1,210/1,224 APA2 events.

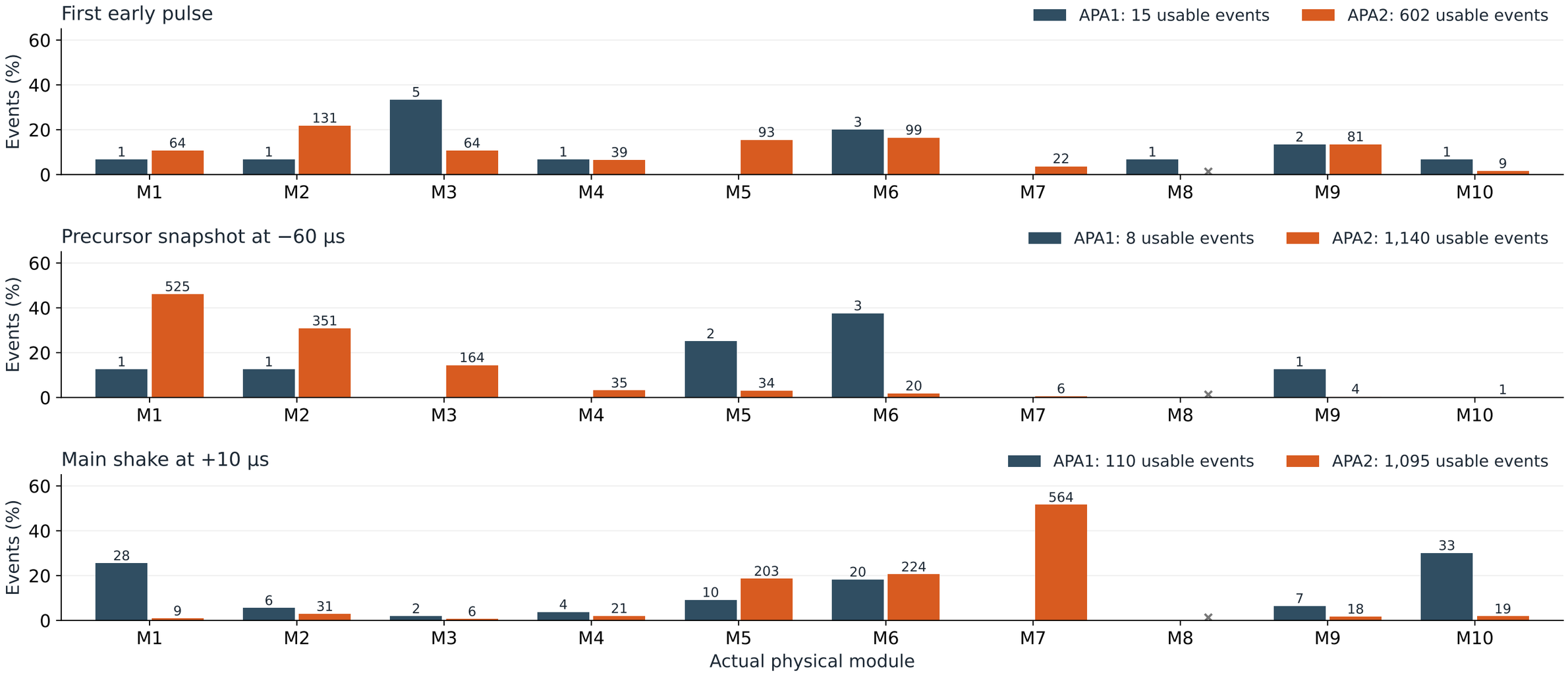
Each bar combines the recorded readouts belonging to that physical module. At least two must be recorded at the same time.

Use the middle fluctuation during −120 to −20 µs, then the middle value across events. n is the number of usable events for that module.

The values measure recorded waveform variation, not calibrated light. Missing recording is not zero activity. October APA1 and November APA2.

The main shake favours different physical modules

Bars count events whose largest recorded signal belongs to that module. M1 is at the top; M10 is at the bottom.



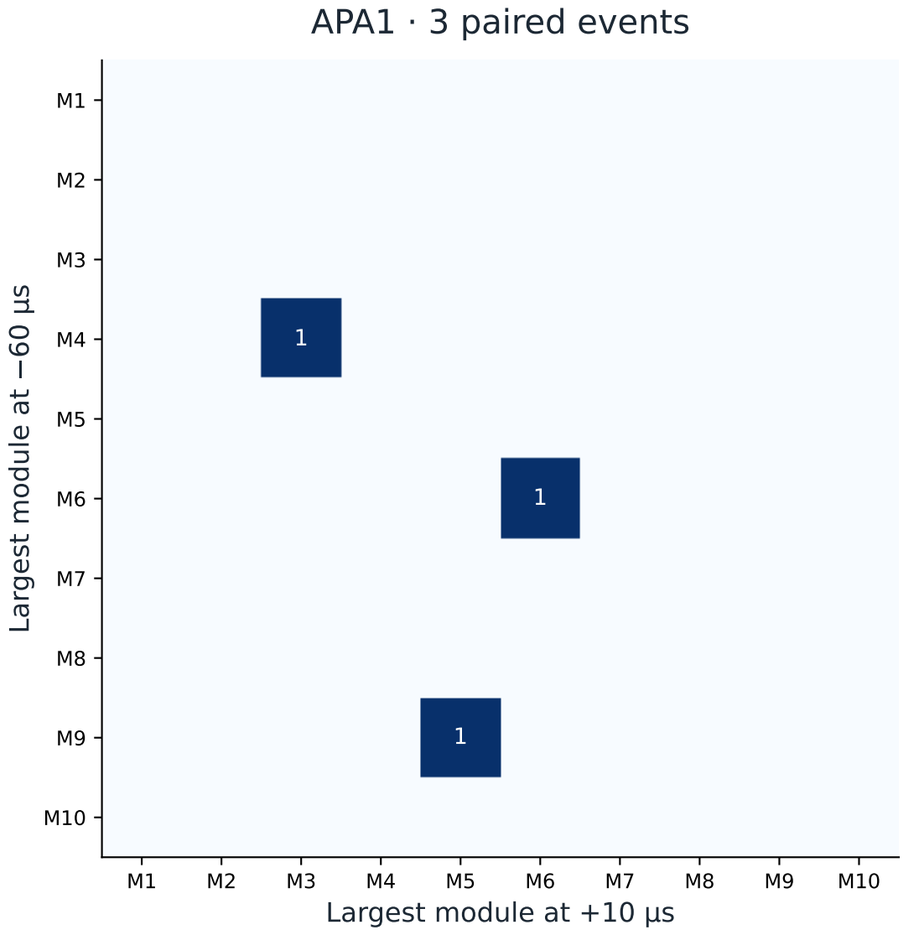
Main shake: APA1 is most often largest in M10, M1 or M6 (81/110 events); APA2 in M5–M7 (991/1,095 events).

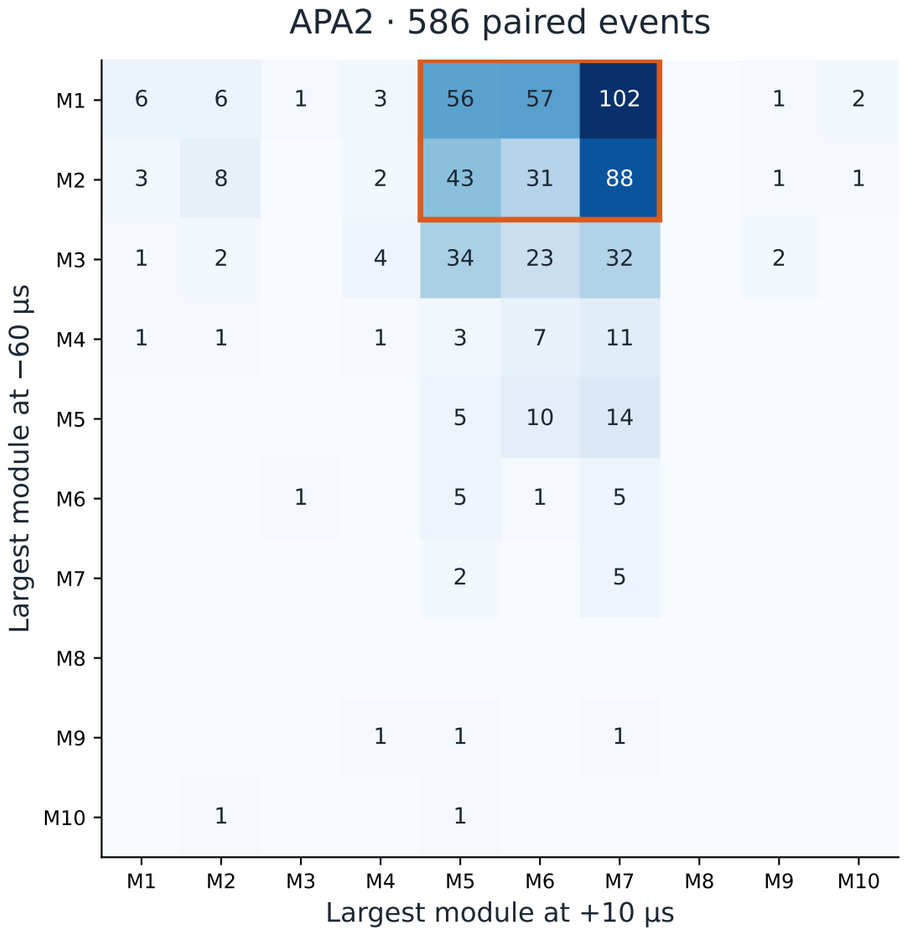
Numbers above bars count events. A module uses its largest recorded input at this instant; this differs from the typical activity on page 1.

APA1 has only 8 usable −60 µs snapshots. Different stages can contain different events. × marks excluded APA2 M8; APA1 M8 remains included.

APA1 has too few paired snapshots to test the same spatial sequence

Compare −60 µs with +10 µs in the same event, using exactly the same recorded channels at both times.





APA2: 377/411 events strongest in M1–M2 at −60 µs become strongest in M5–M7 at +10 µs (92%).

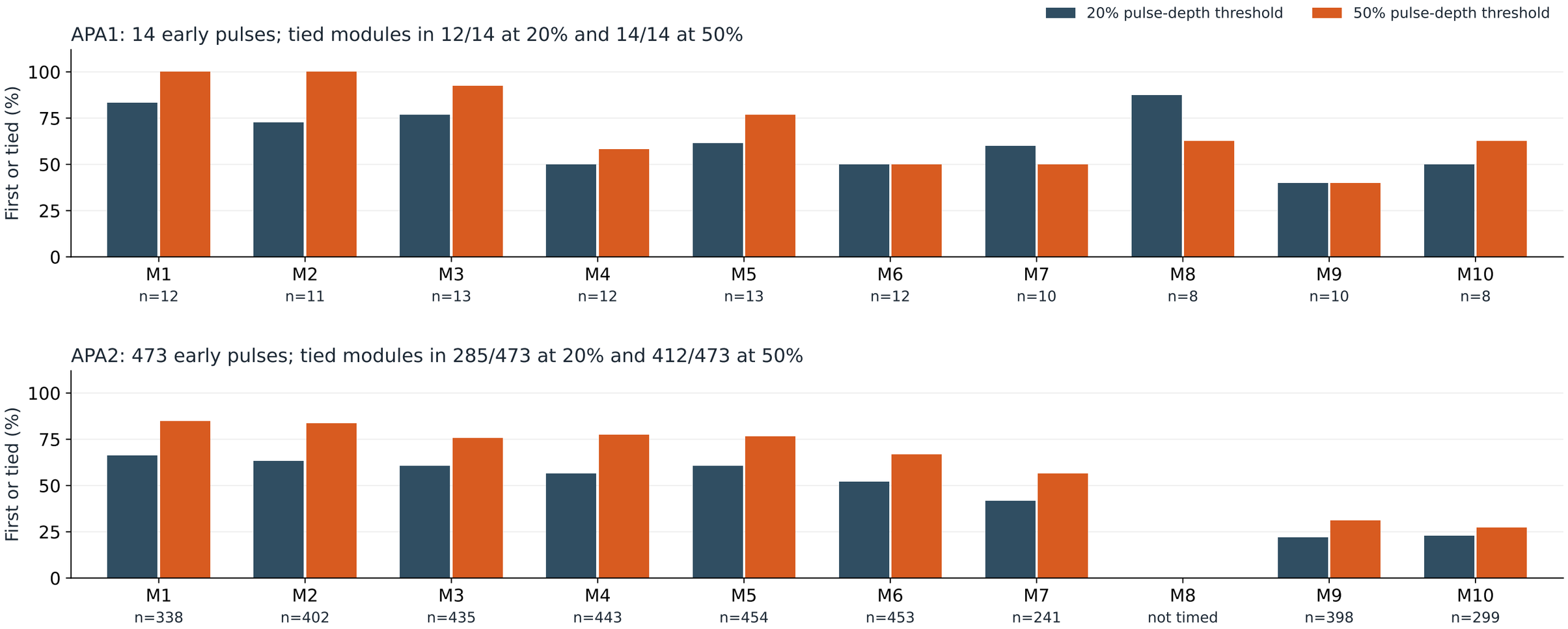
APA1: only 3 events support this test. This is insufficient to establish or exclude a repeatable field-cage-to-centre sequence.

Each cell counts events. These are physical modules. The darker colour means more events within that panel; colour scales differ.

The comparison measures a change in the largest recorded response. It does not measure a travelling source or propagation speed.

Several physical modules start together

The first selected sharp pulse, measured by module. A module needs at least two timed readouts; each event needs at least four timed modules.



APA1 M1, M2 and M3 are often in the earliest group. No APA1 module consistently starts alone under both thresholds.

A module starts at the earliest measured onset among its recorded inputs. Onsets within 16 ns count as tied; n counts measurable pulses.

Requiring a second input in the module to cross the threshold also gives many ties. A 100% bar can mean always tied in the measured subset.

This times sharp early pulses. It does not determine the start of the slow precursor. Instrument delays are not calibrated.